
Word Embeddings on a Variable-Curvature Riemannian Manifold

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Abstract

Hyperbolic embedding offers an efficient method for modeling hierarchical taxonomies in embedding tasks, yet existing paradigms are constrained by assuming fixed constant curvature. This rigid geometric assumption fails to capture the intrinsic topological heterogeneity of lexical semantics. To resolve this structural mismatch, we propose a learnable variable-geometry framework governed by a position-dependent Riemannian metric field. Specifically, we parameterize a neural conformal to empower distinct semantic regions to autonomously adapt their curvature while preserving the space’s hierarchical bias. To ensure computational scalability and stability, we introduce differentiable path-approximation strategies for geodesic distance estimation, coupled with magnitude and smoothness regularizations to prevent metric degeneracy. Our work refines semantic representations through a flexible differential-geometric method, providing a locally adaptive paradigm for manifold learning in NLP.

Keywords: Riemannian manifold; geodesic distance; hyperbolic embeddings; manifold learning; natural language processing

1 Introduction

Word embeddings have become a fundamental representation technique in modern natural language processing. Classical methods such as Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014) represent words as vectors in a Euclidean space, where semantic similarity is usually measured by cosine similarity or Euclidean distance. Later contextual models such as BERT (Devlin et al., 2019) further improve word representations by making them dependent on sentence-level context. These methods have achieved strong empirical success and have shown that many semantic and syntactic regularities can be effectively encoded in continuous vector spaces.

However, a structural limitation of Euclidean representation has long remained under-addressed: Euclidean space is not naturally suited for modeling hierarchical semantic relations (Sarkar, 2011; Nickel and Kiela, 2017; Sala et al., 2018; Tifrea et al., 2019). Many lexical concepts are organized not only by local similarity, but also by taxonomic abstraction and specialization. For example, a simple semantic chain may take the form:

$$\text{entity} \rightarrow \text{animal} \rightarrow \text{mammal} \rightarrow \text{dog}.$$

As the hierarchy becomes deeper, the number of concepts can grow rapidly through branching. In a tree-like structure, this growth is often exponential according to the depth. Nevertheless, the volume of a Euclidean space grows only polynomially with its radius. This creates a fundamental mismatch between the geometry of Euclidean space and the geometry of hierarchical data (Sarkar, 2011; Sala et al., 2018).

This mismatch between exponential and polynomial growth can lead to severe distortion when hierarchical semantic structures are embedded into a flat space. Closely related concepts may be

forced too far apart, while distant concepts from different branches may become artificially close. Alternatively, one may need to increase the embedding dimensionality substantially in order to reduce such distortion. Therefore, the limitation is not simply that Euclidean embeddings are less expressive in practice, but that their globally flat geometry imposes an inductive bias that is poorly aligned with tree-like semantic organization (Ganea et al., 2018; Tifrea et al., 2019).

Existing work has mainly addressed this limitation in two ways. One direction keeps the Euclidean representation space but increases the dimensionality or model capacity (Bourgain, 1985; Mikolov et al., 2013), hoping that a sufficiently high-dimensional flat space can approximate complex semantic relations. However, it would require more calculation complexity and storage. Another direction replaces Euclidean geometry with non-Euclidean geometry, especially hyperbolic geometry. Hyperbolic spaces are particularly suitable for hierarchical representation learning because negatively curved spaces have exponential volume growth, which naturally matches the expansion pattern of trees. Poincaré embeddings demonstrate that hierarchical symbolic data, such as WordNet hypernymy graphs, can be effectively represented in a low-dimensional hyperbolic space (Nickel and Kiela, 2017), as illustrated in Figure 1. Subsequent work has further developed this idea through Lorentz models, hyperbolic entailment cones, hyperbolic word embeddings, and hyperbolic graph neural networks (Ganea et al., 2018; Tifrea et al., 2019; Chami et al., 2019).

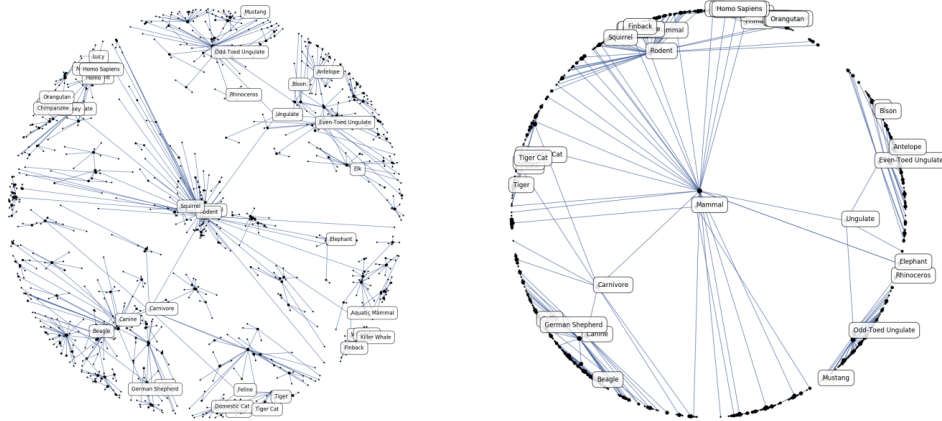


Figure 1: Two-dimensional Poincaré embeddings of the transitive closure of the WordNet mammals subtree. Adapted from (Nickel and Kiela, 2017).

Despite these advances, the above methods only alleviate the flatness problem of Euclidean space and do not fully remove the rigidity of the underlying geometry: These methods still rely on a fixed or regionally fixed curvature structure. Standard Poincaré embeddings assume one fixed negative curvature throughout the embedding space. Lorentz embeddings use a different model of the same constant-curvature hyperbolic geometry. Mixed-curvature product spaces improve flexibility by combining several components of constant-curvature, but the curvature within each component remains fixed (Gu et al., 2019).

Therefore, these methods still impose a globally shared or piecewise fixed geometric bias. This rigidity is problematic for lexical semantics, since it treats all semantic regions as if they followed the same geometric law with the distance growth and hierarchical expansion. In reality, lexical semantics is locally heterogeneous (Hill et al., 2015). Taxonomic regions often contain deep branching structures and may benefit from stronger negative curvature, whereas synonym clusters and local similarity neighborhoods usually exhibit weaker hierarchical expansion and may require a flatter geometry (Tifrea et al., 2019). Context-dependent words further introduce overlapping semantic neighborhoods, making fixed-curvature methods insufficient to provide suitable distance scales for all regions (Miller, 1995; Devlin et al., 2019).

Thus, there remains a fundamental gap between semantic nature and current embedding methods based on globally fixed or piecewise fixed geometry. This motivates a variable-curvature or position-dependent metric, in which different semantic regions can adapt their local geometric properties

according to their own features. In this work, we propose a learnable variable-geometry framework based on a position-dependent Riemannian metric. Instead of assigning one global metric to the entire semantic space, we allow the local geometry to change according to the semantic region. The main research objectives of this work are summarized as follows:

- We propose a word embedding framework with learnable local geometry. Instead of using one fixed metric for the whole space, the model learns a position-dependent Riemannian metric by locally deforming the Poincaré metric, enhancing the representational capacity of the embedding space.
- We define semantic similarity through an approximated geodesic distance under the learned metric, rather than through a globally fixed Euclidean or hyperbolic distance.
- We provide practical differentiable distance approximations and metric regularization terms to make the proposed variable-geometry framework computationally feasible and geometrically stable.

2 Related Work

2.1 Euclidean Word Embeddings and Semantic Representation

Word embeddings have traditionally been learned in Euclidean vector spaces. Representative methods such as Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014) encode words as dense vectors, where semantic similarity is usually measured by cosine similarity or Euclidean distance. These approaches have shown that many lexical regularities can be captured by linear structures in vector space. Later contextual models, such as BERT (Devlin et al., 2019), further improve semantic representation by making word representations dependent on their sentence-level context.

Despite their empirical success, Euclidean embedding spaces are globally flat. This makes Euclidean embeddings less naturally suited for hierarchical or tree-like semantic structures, where the number of concepts can grow rapidly with depth. Prior work on metric embeddings has shown that representing tree-like structures in flat spaces often leads to dimension–distortion tradeoffs (Bourgain, 1985; Sarkar, 2011; Sala et al., 2018). In lexical semantics, this limitation is especially relevant for taxonomic resources such as WordNet, where concepts are connected through hypernymy and hyponymy relations (Miller, 1995). These observations motivate the use of non-Euclidean geometries for hierarchical representation learning.

2.2 Hyperbolic Embeddings for Hierarchical Structures

Hyperbolic geometry has received increasing attention because negatively curved spaces naturally support exponential volume growth, which matches the expansion pattern of tree-like data. Poincaré embeddings (Nickel and Kiela, 2017) are one of the most influential works in this direction. They embed hierarchical symbolic data into the Poincaré ball and show strong performance on WordNet and other hierarchical datasets. Their central insight is that low-dimensional hyperbolic spaces can represent hierarchical structures with lower distortion than comparable Euclidean spaces.

Following this line of work, several studies have explored alternative models and extensions of hyperbolic representation learning. Lorentz embeddings (Nickel and Kiela, 2018) formulate hierarchical representation learning in the Lorentz model of hyperbolic space, improving numerical stability and optimization behavior compared with the Poincaré ball model. Hyperbolic entailment cones (Ganea et al., 2018) extend hyperbolic embeddings by modeling hierarchical relations as partial orders using geodesically convex cones, making them suitable for lexical entailment and directed acyclic graph structures. Poincaré GloVe (Tifrea et al., 2019) further adapts hyperbolic geometry to word embedding learning by combining the GloVe objective with Poincaré geometry, showing that hyperbolic spaces can also be useful for general lexical semantic representation.

These methods demonstrate the advantage of hyperbolic geometry for representing hierarchical data. However, most of them still assume a globally fixed negative curvature. As a result, the entire semantic space is governed by one uniform geometric structure, even though different semantic regions may exhibit different degrees of hierarchy, density, and local semantic organization.

2.3 Non-Euclidean Embeddings for Graphs and Knowledge Representation

Beyond lexical semantics, non-Euclidean embeddings have also been applied to knowledge graphs, multi-relational data, and graph neural networks. Multi-Relational Poincaré embeddings (Balažević et al., 2019) extend hyperbolic representation learning to multi-relational graph data and show that hyperbolic geometry can improve link prediction in hierarchical knowledge graphs. Hyperbolic Graph Convolutional Networks (Chami et al., 2019) introduce graph neural network operations in hyperbolic space and demonstrate advantages on hierarchical and scale-free graphs.

These works suggest that non-Euclidean geometry is useful not only for lexical taxonomies but also for broader relational structures. In particular, graphs with hierarchical or power-law organization can benefit from the geometry of hyperbolic space. Nevertheless, many of these approaches still rely on a predefined hyperbolic model with fixed curvature. Therefore, while they improve over Euclidean methods in representing hierarchical expansion, they do not fully address the problem of locally varying geometric structure.

2.4 Mixed-Curvature and Product Manifold Representations

Another direction aims to increase geometric flexibility by combining multiple geometric components. Mixed-curvature product spaces (Gu et al., 2019) represent data in a combination of constant-curvature manifolds, such as Euclidean, spherical, and hyperbolic components. This allows the embedding space to capture different types of relational structure more flexibly than a single Euclidean or single hyperbolic space.

However, mixed-curvature product spaces normally assign a constant curvature to each component. Although this improves global flexibility, the local geometry within each component remains fixed. Thus, these methods still rely on a component-wise geometric structure rather than a continuously varying curvature field. For lexical semantics, this may be restrictive, because semantic regions cannot always be cleanly separated into a few globally specified geometric components. This suggests the need for a position-dependent Riemannian metric, where distance is determined by a learned local metric field instead of a fixed curvature or a product of fixed-curvature components.

2.5 Riemannian Metric Learning and Variable Geometry

The Riemannian metric allows the distance structure to vary continuously across the embedding space. In a Riemannian manifold, the metric defines an inner product on each tangent space, so local distances can change from one region to another while still preserving a coherent geometric structure. (Lee, 2018)

This perspective is closely related to manifold learning, where data are assumed to lie on or near a nonlinear geometric structure rather than in a globally flat Euclidean space. Recent work has also studied how nonlinear latent spaces can induce non-Euclidean geometry, where distances should be measured according to a local Riemannian metric rather than a fixed Euclidean metric (Arvanitidis et al., 2018).

For semantic representation, this variable-geometry perspective is especially relevant because lexical semantics is locally heterogeneous. Taxonomic regions may require strongly hyperbolic behavior, while synonym clusters or local similarity neighborhoods may be closer to flat geometry. Context-dependent and polysemous words can further create overlapping semantic neighborhoods, where a single global curvature or a small set of fixed-curvature components may not provide suitable distance scales for all regions (Miller, 1995; Hill et al., 2015; Devlin et al., 2019).

Therefore, a learnable position-dependent metric provides a natural way to move from fixed or component-wise geometry toward continuously adaptive semantic geometry. In this work, we follow this direction by locally deforming the Poincaré metric with a neural scalar field. This design preserves the hierarchical inductive bias of hyperbolic geometry while allowing the local distance scale to adapt across different semantic regions.

3 Methodology

This section introduces our variable-geometry embedding framework. The main idea is to extend the fixed-curvature Poincaré embedding space by learning a position-dependent Riemannian metric. Instead of directly assigning a curvature value to each point, we parameterize a smooth metric field over the embedding space. The local curvature is then induced by the learned metric.

Given two word representations, the learned metric is used to define a geometry-aware distance between them. Since the exact geodesic distance under a general position-dependent metric is usually difficult to compute in closed form, we approximate it using differentiable path-based distance estimators, including a midpoint approximation and a multi-segment path approximation. The resulting distance is then used in a pairwise contrastive training objective: semantically or hierarchically related word pairs are encouraged to have smaller distances, while negative pairs are pushed farther apart. In this way, both the word coordinates and the neural metric field are learned through the semantic distance loss.

3.1 Motivation

Poincaré embeddings model hierarchical structures by placing data points inside the Poincaré ball. Let $\mathbb{B}^d = \{x \in \mathbb{R}^d : \|x\| < 1\}$ denote the open unit ball. The standard Poincaré ball is equipped with the conformal metric

$$g_x^{\text{Poincare}} = \lambda_{\text{Poincare}}(x)^2 I, \quad \text{where} \quad \lambda_{\text{Poincare}}(x) = \frac{2}{1 - c\|x\|^2}.$$

Here $c > 0$ denotes the curvature-related parameter and I is the identity matrix. This metric is conformal, meaning that it locally preserves angles while scaling all directions equally. However, the curvature structure of the standard Poincaré ball is globally fixed and may not be flexible enough to model semantic regions with different local geometric properties.

To address this limitation, we introduce a learnable conformal deformation of the Poincaré metric. The goal is to allow different semantic regions to have different local distance scales, while keeping the metric positive definite and computationally stable.

3.2 Embedding Space

Let $w_i \in \mathbb{R}^{d_0}$ be the initial vector representation of word i , for example obtained from Word2Vec. We first map w_i to a coordinate point x_i in the embedding manifold.

In the simplest Euclidean-coordinate version, we may directly use $x_i = w_i$. In the Poincaré-ball version, the point must satisfy $\|x_i\| < 1$. Therefore, we use a learnable projection $z_i = f_\psi(w_i)$, followed by a projection into the unit ball:

$$x_i = \tanh(\|z_i\|) \frac{z_i}{\|z_i\| + \epsilon},$$

where $\epsilon > 0$ is a small constant for numerical stability. This guarantees that $\|x_i\| < 1$. The key point is that the embedding vector x_i is treated as a coordinate point on the manifold. We do not explicitly map the vector through the exponential map. Instead, the Riemannian structure is determined by the metric field defined on the coordinate space.

3.3 Learnable Conformal Metric

We define a neural scalar field $u_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$, where u_θ is implemented by a multi-layer perceptron. This scalar field controls the local deformation of the metric. The proposed variable conformal metric is defined as

$$g_x = \lambda_{\text{Poincare}}(x)^2 \exp(2u_\theta(x)) I.$$

Equivalently,

$$g_x = \left(\frac{2}{1 - c\|x\|^2} \right)^2 \exp(2u_\theta(x))I.$$

This construction has several advantages:

- **Positive definiteness.** The metric is automatically positive definite because $\lambda_{\text{Poincare}}(x)^2 > 0$ and $\exp(2u_\theta(x)) > 0$. Therefore, for any non-zero tangent vector $v \neq 0, v^\top g_x v > 0$.
- **Preservation of hierarchical inductive bias.** The factor $\lambda_{\text{Poincare}}(x)$ comes from the standard Poincaré metric and preserves the hierarchical inductive bias of hyperbolic geometry.
- **Local geometric flexibility.** This allows different semantic regions to have different local distance scales.
- **Conformal structure.** The identity matrix I indicates that the proposed metric is conformal. Therefore, the local geometry rescales tangent-vector lengths while preserving local angles.

The squared norm of a tangent vector $v \in T_x \mathcal{M}$ under this metric is

$$\|v\|_{g_x}^2 = v^\top g_x v = \lambda_{\text{Poincare}}(x)^2 \exp(2u_\theta(x)) \|v\|^2.$$

Thus, the local length is scaled by: $\lambda_{\text{Poincare}}(x) \exp(u_\theta(x))$.

3.4 Distance Approximation

After defining the position-dependent metric, we need a way to compute distances between word representations. This distance is essential for training: positive word pairs should be assigned smaller distances, while negative pairs should be pushed farther apart. Therefore, the distance function connects the learned Riemannian metric with the semantic objective.

For a general position-dependent metric, the natural choice is the geodesic distance. Given two embedded word representations x_i and x_j , it is defined as

$$d_g(x_i, x_j) = \inf_{\gamma} \int_0^1 \sqrt{\dot{\gamma}(t)^\top g_{\gamma(t)} \dot{\gamma}(t)} dt,$$

where $\gamma : [0, 1] \rightarrow \mathcal{M}$ is a smooth curve satisfying $\gamma(0) = x_i$ and $\gamma(1) = x_j$. This distance measures the length of the shortest path between the two points under the learned local metric.

However, the exact geodesic distance is generally not available in closed form when the metric varies with position. Since the metric field changes across the embedding space, the shortest path depends on the local geometry along the entire curve. Computing it would require solving the geodesic equation or optimizing over possible paths, which is expensive and unstable for neural training. Therefore, we approximate the geodesic distance using differentiable path-based estimators. We consider a midpoint approximation and a multi-segment path approximation, both of which can be efficiently used in the contrastive loss.

3.4.1 Midpoint Approximation

The simplest way to approximate the geodesic distance is to assume that the metric does not change significantly between two nearby embeddings. Under this assumption, the local geometry along the path from x_i to x_j can be approximated by evaluating the metric at their midpoint

$$m_{ij} = \frac{x_i + x_j}{2}.$$

Then the distance between x_i and x_j can be approximated by measuring the Euclidean displacement $x_i - x_j$ under the local metric at m_{ij} :

$$d_g(x_i, x_j) \approx \sqrt{(x_i - x_j)^\top g_{m_{ij}} (x_i - x_j)}.$$

Substituting the proposed conformal metric gives

$$d_g(x_i, x_j) \approx \lambda_{\text{Poincare}}(m_{ij}) \exp(u_\theta(m_{ij})) \|x_i - x_j\|.$$

This approximation is computationally efficient because it requires only one evaluation of the learned metric field for each word pair. It is also fully differentiable, so gradients can be propagated through both the word embeddings and the neural scalar field u_θ . However, the midpoint approximation is most reliable when the metric varies smoothly between x_i and x_j . If the local geometry changes sharply along the path, evaluating the metric only at the midpoint may ignore important variations in the learned distance scale.

3.4.2 Multi-Segment Path Approximation

To obtain a more accurate estimate, we further approximate the path length by evaluating the metric at multiple points along the path between x_i and x_j . Instead of representing the whole path by a single midpoint, we discretize the straight line segment connecting the two embeddings. Let

$$\gamma_t = \left(1 - \frac{t}{T}\right) x_i + \frac{t}{T} x_j, \quad t = 0, \dots, T,$$

where T is the number of segments. The displacement of each small segment is given by $\gamma_{t+1} - \gamma_t$. The geodesic distance is then approximated by the accumulated local path length:

$$d_g(x_i, x_j) \approx \sum_{t=0}^{T-1} \sqrt{(\gamma_{t+1} - \gamma_t)^\top g_{\gamma_t} (\gamma_{t+1} - \gamma_t)}.$$

Using the conformal metric, this becomes

$$d_g(x_i, x_j) \approx \sum_{t=0}^{T-1} \lambda_{\text{Poincare}}(\gamma_t) \exp(u_\theta(\gamma_t)) \|\gamma_{t+1} - \gamma_t\|.$$

This approximation allows the model to account for changes in the local metric along the path, rather than relying on a single metric evaluation at the midpoint. As T increases, the approximation can capture finer variations of the learned geometry, but it also requires more evaluations of the neural scalar field. Therefore, T controls a trade-off between computational efficiency and distance accuracy. In practice, a small value of T can already provide a more informative approximation than the midpoint method while remaining efficient enough for neural training.

3.5 Training Objective

We train the model using positive and negative word pairs. Let (i, j^+) denote a positive pair, where words i and j^+ are semantically related or hierarchically connected. Let $\mathcal{N}(i)$ denote a set of negative samples for word i .

The model is trained to minimize the distance between positive pairs and maximize the distance between negative pairs. We use a contrastive softmax loss:

$$\mathcal{L}_{\text{pair}} = -\log \frac{\exp(-d_g(x_i, x_{j^+}))}{\exp(-d_g(x_i, x_{j^+})) + \sum_{k \in \mathcal{N}(i)} \exp(-d_g(x_i, x_k))}.$$

Here $d_g(\cdot, \cdot)$ is the approximated geodesic distance defined above.

The neural metric field $u_\theta(x)$ does not require direct supervision. Its parameters are updated through the pairwise distance loss. The gradient path is $\mathcal{L}_{\text{pair}} \rightarrow d_g \rightarrow u_\theta \rightarrow \theta$. For example, under the midpoint approximation:

$$d_g(x_i, x_j) \approx \lambda_{\text{Poincare}}(m_{ij}) \exp(u_\theta(m_{ij})) \|x_i - x_j\|.$$

Therefore,

$$\frac{\partial d_g(x_i, x_j)}{\partial u_\theta(m_{ij})} = d_g(x_i, x_j).$$

This means that if the loss encourages a pairwise distance to decrease, the gradient can reduce the local scale factor $\exp(u_\theta)$. Conversely, if the loss encourages a pairwise distance to increase, the gradient can increase the local scale factor.

3.6 Regularization of the Metric Field

A freely learned metric field may lead to degenerate solutions. For example, $u_\theta(x)$ may become extremely negative in regions containing positive pairs, causing distances to collapse, or extremely positive in regions containing negative pairs, causing numerical instability. Therefore, we introduce regularization terms.

First, we restrict the output range of the scalar field by defining $u_\theta(x) = a \tanh(h_\theta(x))$, where $h_\theta(x)$ is the raw output of the MLP and $a > 0$ controls the maximum deformation magnitude. This ensures $u_\theta(x) \in [-a, a]$.

Second, we penalize large metric deviations: $\mathcal{L}_{\text{mag}} = \mathbb{E}_x [u_\theta(x)^2]$. This encourages the learned geometry to stay close to the original Poincaré metric unless local deformation is necessary.

Third, we encourage smoothness of the metric field: $\mathcal{L}_{\text{smooth}} = \mathbb{E}_x [\|\nabla_x u_\theta(x)\|^2]$.

This prevents abrupt changes in the metric and encourages a smooth geometric structure.

The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{pair}} + \alpha \mathcal{L}_{\text{mag}} + \beta \mathcal{L}_{\text{smooth}},$$

where $\alpha \geq 0$ and $\beta \geq 0$ are regularization coefficients controlling the strength of the magnitude regularization and smoothness regularization, respectively.

4 Datasets

To evaluate the proposed variable-geometry word embedding framework, we focus on lexical datasets with explicit or graded semantic hierarchy. Since our method is designed to extend fixed-curvature hyperbolic embeddings with a learnable position-dependent metric, hierarchical lexical resources are more suitable than general raw text corpora for the first-stage evaluation.

4.1 WordNet Hypernymy Graph

Our primary training and evaluation data is derived from WordNet, a large lexical database of English in which words are grouped into synsets and connected through semantic relations such as synonymy, hyponymy, and hypernymy (Miller, 1995). Following the experimental setting of Poincaré embeddings (Nickel and Kiela, 2017), we use the noun hypernymy graph as the main hierarchical structure.

Each node in the graph corresponds to a WordNet synset, and each directed edge represents a hypernymy relation. For example, a specific concept such as *dog* can be connected to more general concepts such as *canine*, *carnivore*, and *mammal*. These relations naturally form a hierarchical graph, making WordNet suitable for evaluating whether the learned embedding space can capture both semantic similarity and hierarchical abstraction.

We consider two WordNet-based settings:

- **WordNet Mammals Subtree.** This subset is used as a small-scale proof-of-concept experiment. It provides a compact hierarchical structure that is easy to train, debug, and visualize. In particular, it allows us to inspect whether the learned scalar metric field $u_\theta(x)$ produces meaningful local geometric variations across different semantic regions.
- **WordNet Nouns Hypernymy Graph.** This setting is used as the main evaluation benchmark. It follows the standard setup used in prior work on Poincaré embeddings (Nickel and Kiela, 2017). Compared with the mammals subtree, the full noun hierarchy is larger and more structurally complex, making it suitable for testing whether the proposed variable-geometry metric improves over fixed-curvature baselines.

Positive training pairs are constructed from observed hypernymy edges or their transitive closure. Given a positive pair (i, j^+) , where j^+ is a hypernym or semantically related ancestor of i , negative samples are drawn from nodes that are not directly connected to i in the hierarchy. The model is

trained to assign smaller distances to positive pairs and larger distances to negative pairs under the learned Riemannian metric.

4.2 SimLex-999

As an optional supplementary benchmark, we use SimLex-999 to evaluate general semantic similarity (Hill et al., 2015). SimLex-999 contains 999 word pairs annotated for genuine semantic similarity, explicitly distinguishing similarity from broader association or relatedness.

Although our method is mainly designed for hierarchical semantic structures, SimLex-999 provides a useful additional test of whether the learned variable metric also improves general word similarity estimation. Similar to the HyperLex evaluation, we use the negative learned distance

$$s(w_i, w_j) = -d_g(x_i, x_j)$$

as the model-predicted similarity score and compute Spearman’s rank correlation with human ratings.

4.3 Evaluation Protocol

We evaluate the proposed method under three complementary settings. First, we conduct small-scale controlled experiments on the WordNet Mammals subtree to verify the feasibility of the learned metric field and to visualize local geometric deformation. Second, we evaluate on the full WordNet Nouns hypernymy graph using reconstruction and link prediction metrics, following prior work on hyperbolic representation learning (Nickel and Kiela, 2017). Third, we use HyperLex and optionally SimLex-999 as external evaluation benchmarks to test whether the learned distances align with human judgments of lexical entailment and semantic similarity.

This design allows us to isolate the contribution of the learned position-dependent metric from the effect of hyperbolic geometry itself.

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